Riley Loo

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Experience

Member of Technical Staff Jan 2026 - Jul 2026
Bold Software Inc *San Francisco, CA*

- Minimized downtime for most active users during the Januscape CVE response by implementing QoS priority for bulk suspend/resume operations
- Optimized VM disk and RAM state transfer functions, allowing for multi-gigabit transfers between London and New York, unlocking inter-hypervisor transfers for load-balancing and downtime mitigation purposes
- Implemented vital user-facing control features such as external restart, fast VM cloning leveraging ZFS Copy on Write, intra-organization and public access controls

Software Engineering Intern August 2025 - Jan 2026
Fyra Labs LLC *Minneapolis, MN*

- Maintained packages on the Terra package repository for Fedora and RHEL
- Assisted in maintaining a Fedora-based distribution, foucsing on user experience and GUI responsiveness
- Implemented a Nix installer for Ultramarine Linux ahead of the official FESCo Nix directory decision

FFmpeg Code Contributor May 2023 - August 2023
Google Summer of Code *Mountain View, CA*

- Implemented VVC (H.266) decoding in-loop filter kernels in assembly with AVX2 SIMD to maximize parallelism
- Achieved a 3.6% speedup compared to compiler output, achieving smooth 4K playback on commodity hardware, preparing FFmpeg for next-generation video codecs
- Employed GitHub Actions and TDD principles to write optimized `x86_64` assembly

System Administrator May 2023 - May 2025
University of Minnesota ACM Student Chapter *Minneapolis, MN*

- Administered and provisioned over 100 virtual machines for community use using `bhyve` and KVM
- Managed a diverse fleet of servers, from ZFS storage servers to multi-GPU systems for LLM inference, stimulating student interest in Unix and computer networking skills
- Overhauled the ACM UMN network with symmetric 10 Gigabit and 256 publicly addressable IPv4 addresses with Cisco and Arista switches and OPNSense firewall

Education

University of Minnesota Twin Cities Minneapolis, MN
Master of Science, Computer Science *Sep 2024 - May 2025*

- Selected Coursework: **Real-Time and Embedded Operating Systems**, **Error Correcting Codes**, **Advanced Computer Architecture**, Computer Graphics, Programming Languages Theory

Selected Projects

badge-gb | *C*, *CMake*, *Pi Pico SDK*, *RP2040/RP2350*

- A GameBoy emulator for the RP2350-based badge from Github
- Implemented PIO ASM routines for the ST7789 screen over a parallel 8080 bus in a blocking DMA design
- Employed embedded multi-core programming techniques to achieve efficient emulation of GameBoy games

OPL3Duo! VGM player | *C++, MPLAB X, AVR, SPI, I2C*

- Designed and implemented a RTOS-like executive for I/O heavy routines
- Ported the OPL3Duo! `.vgm` playback code from Arduino to the AVR-BLE Development Board
- Modified a C-based SD card driver and FAT32 file system implementation to read data from an SD card over SPI efficiently

Technical Skills

Languages: C, C++, Rust, OCaml, Go, TypeScript, Python, Haskell, Java, `x86_64` Assembly
Frameworks: Ember, Svelte, Flask, Zola, GCP (Firebase, Firestore), Vue
Developer Tools: Git, Nix, Docker / Podman, OpenMPI, CUDA, Bash, KVM, LVM, ClickHouse, gRPC
Libraries: PyTorch, pandas, NumPy, Matplotlib, D3, zlib, imgui, libavcodec, SQLite, PostgreSQL, ZFS
Operating Systems: Debian, Ubuntu, NixOS, Fedora, RHEL, Rocky Linux, macOS